

<d> Add Round key:-

In add round key, the output of mix column transformation is XOR ($\oplus$) with the round key of the corresponding column.

Asymmetric Cryptographic Technique:-

→ Use of separate keys for encryption & decryption.

→ Example is RSA (Rivest Shamir Adleman's) Algorithm.

RSA Algorithm (Block ciphering Technique)

→ Based on the concept of prime number, larger the prime number, more secure will be the data.

Steps of RSA Algorithm:-

<1> Select 2 prime numbers p and q .

<2> Compute $n = p \times q$

<3> Calculate Euler's Totient Function $\phi(n) = (p-1)(q-1)$.

<4> Compute (choose) value of e such that $1 < e < \phi(n)$ and $\gcd(\phi(n), e) = 1$.

<5> Calculate $d \equiv e^{-1} \pmod{\phi(n)} \Rightarrow (d \times e) \pmod{\phi(n)} = 1$

<6> Public key:- $\{e, n\}$

Private key:- $\{d, n\}$

Encryption

$$C = M^e \pmod{n}$$

M = no. of letters in PT

C = no. of letters in CT

Decryption

$$M = C^d \pmod{n}$$

→ Public key is sharable through out the network, but private key is always kept as a secret key and is used for basically decryption.

Q Calculate the value of public key & private key if the prime numbers are given as 3 & 11.

Ans:- <1> $p = 3, q = 11$

<2> $n = p \times q$
 $= 3 \times 11$
 $= 33$

<3> $\phi(n) = (p-1)(q-1)$
 $= (3-1)(11-1)$
 $= 20$

<4> $e = 7$ on 3
 $e < \phi(n)$ and $\gcd(7, 20) = 1$

<5> $(d \times e) \bmod 20 = 1$
 $\Rightarrow (d \times 7) \bmod 20 = 1$
 $d = 3$

The modulo inverse for larger prime number can be computed by extended Euclidean Algorithm.
values of

<6> Public key $\{7, 33\}$
Private key $\{3, 33\}$

Q Show the encryption & decryption for the above computed public & private key if $M=4$.

Encryption:- $C = M^e \bmod n$
 $= 4^7 \bmod 33$
 $= 16$

Decryption:-
 $M = C^d \bmod n$
 $= 16^3 \bmod 33$
 $= 4$

$$\begin{cases} 4 \bmod 33 = 4 \\ 4^2 \bmod 33 = (4 \bmod 33 \times 4 \bmod 33) = 16 \\ 4^4 \bmod 33 = 4^2 \bmod 33 \times 4^2 \bmod 33 = (16 \times 16) \bmod 33 = 256 \bmod 33 = 25 \\ 4^7 \bmod 33 = 4^4 \bmod 33 \times 4^2 \bmod 33 \times 4 \bmod 33 \\ = (25 \times 16 \times 4) \bmod 33 \\ = 16 \end{cases}$$

$16 \bmod 33 = 16$

$16^2 \bmod 33 = 16 \times 16 = 256 \bmod 33 = 25$

$16^3 \bmod 33 =$

Q calculate the values of private & public key if
 $p=5, q=11$. Also compute no. of character in CT if
 $M=9$.

Ans: 1 $p=5, q=11$

<2> $n = 5 \times 11 = 55$

<3> $\phi(n) = (5-1)(11-1)$
 $= 4 \times 10$
 $= 40$

<4> $e = 3$ or 37

$\gcd(3, 40) = 1$

<5> $(d \times e) \bmod \phi(n) = 1$

$\Rightarrow (d \times 3) \bmod 40 = 1$

$\Rightarrow d = 27$

$(3 \times d) \bmod 40 = 1$

$$\begin{array}{r} 41 \\ 81 \\ 121 \\ 161 \end{array} \quad d = 27$$

<6> Public key - $\{3, 55\}$

Private key - $\{27, 55\}$

Encryption:-

$C = M^e \bmod n$
 $= 9^3 \bmod 55$
 $= 14$

$9 \bmod 55 = 9$

$9^3 \bmod 55 = (9 \times 9) \bmod 55 = 81 \bmod 55 = 2$

Decryption:-

$M = C^d \bmod n$
 $= 14^{27} \bmod 55$
 $= 9$

Modular Exponentiation:-

<a> $31^4 \bmod 30$

$31 \bmod 30 = 1$

$31^4 \bmod 30 = 31 \bmod 30 \times 31 \bmod 30 \times 31 \bmod 30 \times 31 \bmod 30$
 $= 1 \times 1 \times 1 \times 1$
 $= 1$

 $98^4 \bmod 30$

$98 \bmod 30 = 28$ or (-2)

$98^4 \bmod 30 = 28^4 \bmod 30 = (-2)^4 \bmod 30$
 $= 28 \bmod 30$
 $= 28 \times 28 \bmod 30 = 28 \times 28 = 784$
 $784 \bmod 30 = 28 \bmod 30$
 $= 28 \times 28 \times 28 \times 28 = 614656$

$$(c) 978^{974} \bmod 979 =$$

$$\underline{978} \bmod \underline{979} = 978 \text{ or } (-1)$$

$$978^{974} \bmod 979 = (-1)^{974} \bmod 979 = 1$$

~~278~~

$$(b) 28^4 \bmod 30 =$$

$$\underline{28} \bmod \underline{30} = 28 \text{ or } (-2)$$

$$28^2 \bmod 30 = 28 \bmod 30 \times 28 \bmod 30$$

$$= (-2 \times -2) = 4$$

$$-2 \bmod 30$$

$$= 30 - 2 \bmod 30$$

$$= 28$$

$$(28 \equiv -2)$$

$$28^4 \bmod 30 = 28^2 \bmod 30 \times 28^2 \bmod 30$$

$$= 4 \times 4$$

$$= 16$$

$$\underline{\underline{Q}} \quad 978^{975} \bmod 979$$

$$(-1) \bmod 979 = 978$$

$$(b - a \bmod a)$$

Calculate:-

$$(1) 29^{107} \bmod 30$$

$$(2) 11^7 \bmod 13$$

$$(3) 3^{100} \bmod 29$$

$$(1) 29^{107} \bmod 30$$

$$\underline{29} \bmod \underline{30} = 29 \text{ or } (-1)$$

$$29^{107} \bmod \underline{30} = (-1)^{107} \bmod \underline{30} = -1 \bmod 30$$

$$= 29$$

$$(2) 11^7 \bmod 13$$

$$\underline{11} \bmod \underline{13} = 11 \text{ or } (-2)$$

$$(-2)^7 \bmod 13 = -128 \bmod 13$$

$$11^2 \bmod 13 = 4$$

$$11^4 \bmod 13 = 16 \bmod 13 = 3$$

$$11^7 \bmod 13 = 11^4 \bmod 13 \times 11^2 \bmod 13 \times 11 \bmod 13$$

$$= 3 \times 4 \times (-2)$$

$$= -24 \bmod 13 = (13 - 24 \bmod 13) = (13 - 11) = 2$$



$$23 > 3^{100} \bmod 29$$

$$3 \bmod 29 = 3$$

$$3^2 \bmod 29 = 9$$

$$3^4 \bmod 29 = 81 \bmod 29$$

$$= 23 \text{ (or } -6)$$

$$3^8 \bmod 29 = 36^2 \bmod 29$$

$$= 7$$

$$3^{16} \bmod 29 = 49 \bmod 29$$

$$= 20 \text{ or } -9$$

$$3^{32} \bmod 29 = 81 \bmod 29$$

$$= -6$$

$$3^{64} \bmod 29 = 36 \bmod 29$$

$$= 7$$

$$3^{100} \bmod 29 = 3^{64} \bmod 29 \times 3^{32} \bmod 29 \times 3^4 \bmod 29$$

$$= (7 \times (-6) \times (-6)) \bmod 29$$

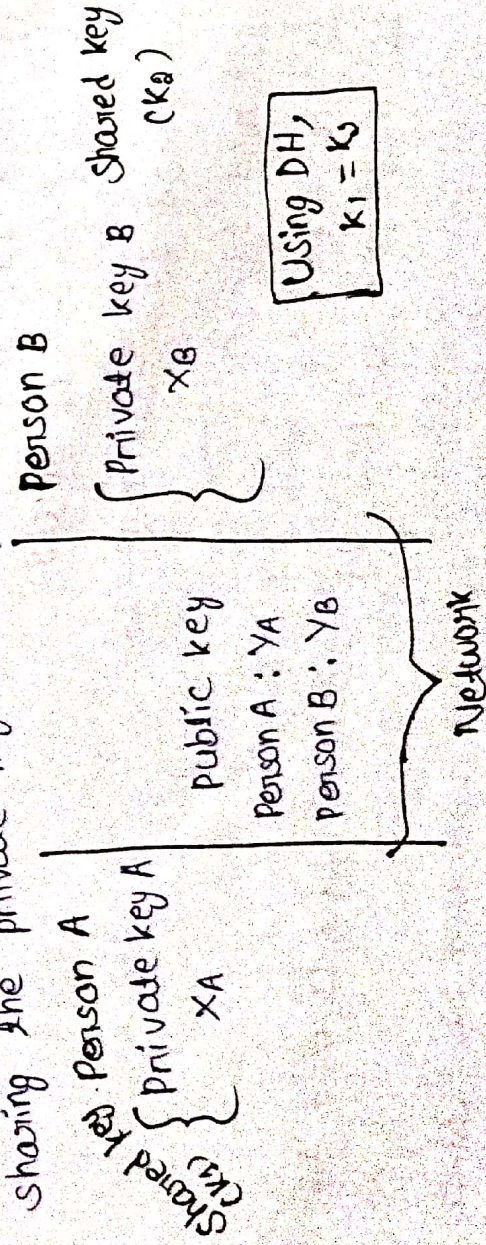
$$= 252 \bmod 29$$

$$= 20$$

Diffie-Hellman's Algorithm

→ This algorithm is a method used in cryptography to security exchange & secret key over an insecure network.

→ The basic idea here is that two persons create a shared secret key using public key & private key without able sharing the private key directly.



→ This algorithm is a key exchange algorithm that uses asymmetric encryption technique for generating a shared key that is computed by both Person A & Person B & happen to be similar.

→ Steps of the Algorithm:-

Step-1:- We have to assume a large prime no. P and its generator g that is known to both person A & person B and values are public.

Step-2:- Both the person's randomly select themselves with the private key X_A & X_B .

Step-3:- Calculate the public keys of A & B as $Y_A = g^{X_A} \text{ mod } P$

$$Y_B = g^{X_B} \text{ mod } P$$

the shared secret key.

Step-4:- Calculation of $Y_A \text{ mod } P$ } $K_1 = K_2$
For Person A, $K_1 = Y_B \text{ mod } P$
 $K_2 = Y_A \text{ mod } P$

Q1 Calculate the value of shared secret key, if the prime number, given as 7 & private key for both the person's are 3 & 4 respectively.

Ans:- Step-1:- $P = 7$ can not be a generator

Always $g < P$ { 1, 2, 3, 4, 5, 6 }

$$g^1 \text{ mod } 7 = 1 \text{ mod } 7 = 1$$

$$g^2 \text{ mod } 7 = 4 \text{ mod } 7 = 4$$

$$g^3 \text{ mod } 7 = 1$$

$$g^4 \text{ mod } 7 = 2 \times 2 \text{ mod } 7 = 4$$

$$4^1 \text{ mod } 7 = 4$$

$$4^2 \text{ mod } 7 = 2$$

$$4^3 \text{ mod } 7 = 1$$

$$4^4 \text{ mod } 7 = 4$$

$$g = 3 \text{ or } 5$$

$$3^1 \text{ mod } 7 = 3$$

$$3^2 \text{ mod } 7 = 2$$

$$3^3 \text{ mod } 7 = 6$$

$$3^4 \text{ mod } 7 = 4$$

$$6^1 \text{ mod } 7 = 6$$

$$6^2 \text{ mod } 7 = 1$$

$$3^5 \text{ mod } 7 = 5$$

$$3^6 \text{ mod } 7 = 3^5 \times 3^1 = 5 \times 3 = 15 \text{ mod } 7 = 1$$

because it repeated.

Step-2 :- (Private key)

$$X_A = 3 \text{ \& } X_B = 4$$

Step-3 :- (Public key)

$$Y_A = g^{X_A} \text{ mod } P$$

$$= 3^3 \text{ mod } 7$$
$$= 6$$

$$Y_B = g^{X_B} \text{ mod } P$$

$$= 5^4 \text{ mod } P$$
$$= 8$$

Step-4 (Secret key)

$$K_1 = Y_B^{X_A} \text{ mod } P$$

$$= 8^3 \text{ mod } 7$$
$$= 1$$

$$K_2 = Y_A^{X_B} \text{ mod } P$$

$$= 6^4 \text{ mod } P$$
$$= 1$$

$$\therefore K_1 = K_2$$

Q2 Calculate the values of secret key if $P = 23$, $g = 5$, $X_A = 6$, $X_B = 15$.

Ans :- Step-1 :- $P = 23$
 $g = 5$

Step-2 :- $X_A = 6$, $X_B = 15$

Step-3 :-

$$Y_A = g^{X_A} \text{ mod } P$$

$$= 5^6 \text{ mod } 23$$
$$= 10$$

$$Y_B = g^{X_B} \text{ mod } P$$

$$= 5^{15} \text{ mod } P$$
$$=$$

$$5^1 \text{ mod } 23 = 5$$

$$5^2 \text{ mod } 23 = 2$$

$$5^3 \text{ mod } 23 = 10 \text{ mod } 23$$
$$= 10$$

$$5^4 \text{ mod } 23 = 4 \text{ mod } 23$$
$$= 4$$

$$5^5 \text{ mod } 23 = 20 \text{ mod } 23$$
$$= 20$$

$$5^{15} \text{ mod } 23 = 20 \times 4 \times 10 \times 20$$
$$= 160 \times 10 \times 10$$
$$= 1600 \text{ mod } 23$$

Mitigation Techniques For

Chapter-9 Securing a Network

Dt:- 29/04/26

- Symmentation
- Access Control
- Monitoring
- Encryption
- Patching

→ Mitigation techniques help in securing a network by using 5 different applications.

→ Segmentation is the process of sub-dividing a bigger n/w by creating isolated segment that will restrict the movement of attackers. It is divided into physical segmentation, virtual segmentation, subnetting & microsegmentation.

→ Physical segmentation sub-divides a network using router, switch & Firewall, where as virtual segmentation is done by VLAN (Virtual LAN) that creates logical segments within a single switch. Subnetting divides and input network into smaller subnets using different subnet masks. Microsegmentation happens at a granular level in cloud environment and the advantages of segmentation are enhanced security, performance optimization, isolation of critical system, access control and scalability and agility.

→ Access Control is the process of permitting or restricting access to different parts of the network using 3 factors.

<a> ACL (Access Control List) :-

This list is generated by network router which helps in permitting ^{or} denying networking traffic using a set of rules. ACL can be specifically generated for

files and folders, which restricts access to different data, another ACL is for incoming network traffic that has permission to modify the network.

(b) Network Permissions:-

→ It includes some permit or deny operation with the list of applications to modify the existing network.

→ Every application has a specific permit list and a deny list, the deny list prevents applications from running in the system.

→ Monitoring:-
It ensures that no deviation happens in the network by creating a list of log activity, traffic pattern and system behaviour.

→ Encryption:-

It's purpose is to transform sensitive data to unreadable format by using ^{the} CIA concept of security.

→ Patching:-

It is the process of updating software, operating system or firmware to fix bugs, errors, weaknesses or vulnerabilities that attackers could exploit.

→ A patch is a small piece of code released by developers that helps in correcting error, fix security vulnerability & provide improve performance and security.

Security Implications in different Chapter-10 Architectural Security Model

- There exist different architectural model in a n/w for implementing n/w security.
- It is essential for protecting data & resources by creating a multi layer strategy with the key elements firewall, ACL, IDS, IPS and SIEM.
- The firewall is the first line of defence that monitors, & filters incoming & outgoing n/w traffic sometimes automatically & sometimes based on the provision of Access Control list (ACL).
- ACL are the network barrier that permits or blocks specific IPs depending on a criteria. The task of intrusion detection system & intrusion prevention system is to respond to security threats happening in real-time using the SIEM software. (Security, Information and Event management).
- This software collects and analyzes the data from different network resources to identify security threat.

Architecture and Infrastructure N/w concepts:-

- Most of the devices in a network either depend on wireless services or cloud services due to which they are vulnerable to different kinds of cryptographic attack. Therefore, several architecture and infrastructures specific to cloud computing are necessary.
- Cloud computing is a flexible & scalable technology helping in accessing storage of data & applications over internet. There exist 4 different cloud models i.e public, private

hybrid & community model.

- Depending on ^{the} security requirements the most preferable cloud model is the hybrid model or community model.
- Depending on n/w architecture systems are classified into 3 categories:

<1> IaaS (Infrastructure as a System)

<2> PaaS (Platform as a System)

<3> SaaS (Services as a System)

Out of this IaaS is the most preferable architecture as it uses the system of IaC (Infrastructure as a Code)

IaC is the process of defining and managing IT infrastructure using machine readable code that is written in JSON (JavaScript Object Notation) & YAML (Y type Markup Language).

- The benefits of IaC are:- redefined efficiency, consistency and reproducibility, version control & collaboration & existence of different tools & cloud providers like AWS, Azure & Google cloud.

Serverless System:-

- It is a cloud computing model where developers can generate and run applications without managing infra.
- The common examples are different cloud providers which incorporate BaaS (Backend as a Service) architecture which provides all backend operations like authentication & file storage monitor by a third party software CSP (Cryptographic Service Provider),

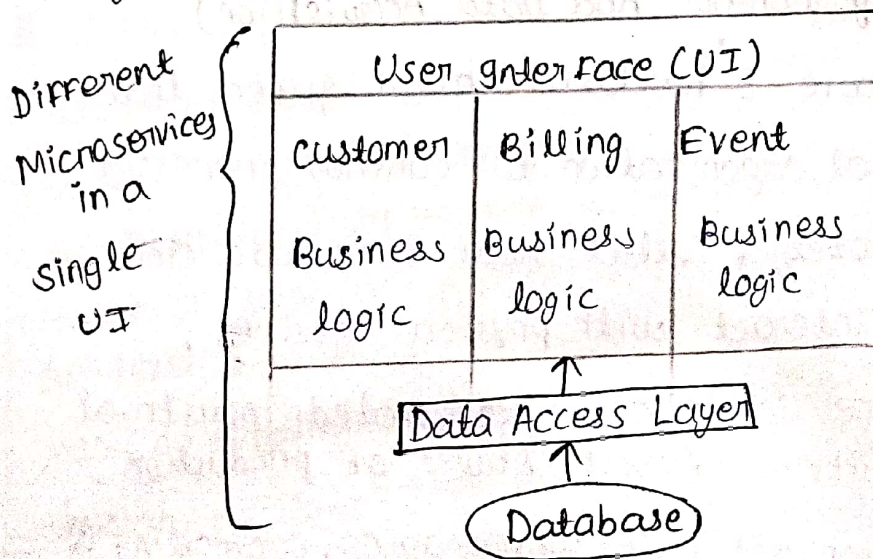
Microservices:-

→ Microservice architecture involves breaking down an existing application into collection of smaller self sufficient services communicating with each other using APIs (Application Programming Interface).

Advantages of microservices:-

→ Agility, scalability, faster development, easy maintenance, improve fault tolerance & system independency.

→ The block structure of a microservice architecture is given as follows:



Internet of Things (IoT):-

→ It refers to the vast network of interconnected physical devices that are embedded with software & connectivity capability & these devices can collect & exchange data with each other over internet.

Due to high amount of data, IoT suffers from many security issues

(a) Lack of standardization:- IoT lacks a standardized security framework making it challenging to ensure proper security criteria.

(b) Data privacy Concerns:- The interconnected devices can mishandle large amount of data available.

(c) Insecure Communication:- IoT devices communicate over insecure channel.

(d) Lifecycle Management:- IoT devices may not have security update or support feature making them vulnerable to security flaws.

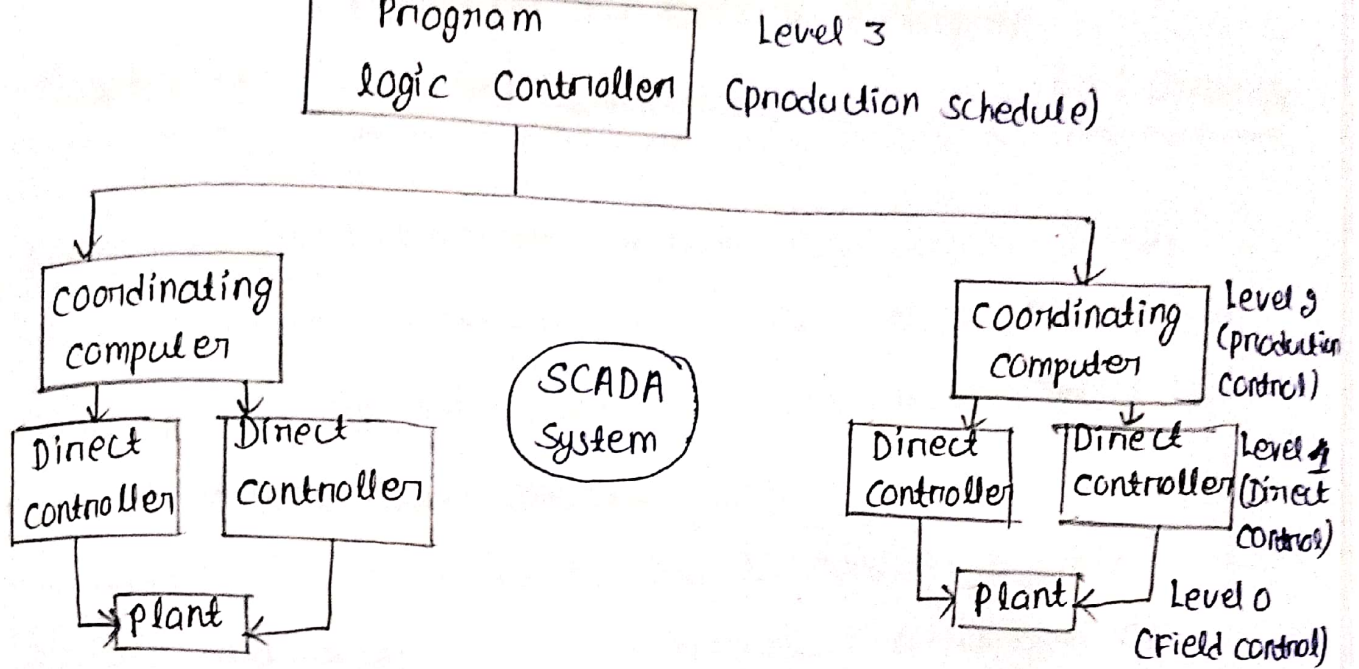
(e) IoT devices can suffer from physical attacks due to supply risk & lack of user awareness.

gmp
SCADA (Supervisory Control And Data Acquisition)

→ It is a software & hardware based system that allows industrial organization to control processes locally or remotely, collect real time data and simultaneously interact with physical devices.

→ These systems are sophisticated, automated industrial control system with different stages of production.

→ They play an important role in managing, controlling & monitoring industrial processes. The block structure is as follows:



→ A SCADA system has 4 phases of operation.

(a) Plant level:- that is level 0 which includes controlling physical devices and equipment.

(b) Controller level:- that is responsible for real time control of physical process using PLC. It is also called as level 1 SCADA.

(c) Coordinating Controller level (level 2):- that uses human machine interface for supervising operation.

(d) Level 3 is the program logic controller level that manages and controls overall production process. SCADA systems are prime target to cyber crime because of certain parameters like usage of energy, manufacturing, logistics & industrial involvement.

Principles to Secure Enterprise

Chapter-11

Infrastructure

Date: 09/05/26

→ Enterprise Infrastructure includes n/w component like router, switches, servers, wireless system & cloud resources. Securing this Infrastructure is essential to protect data & prevent cyber attack.

→ To ensure the security, one has to consider specific scenario and sections of infrastructure which are as follows:

<a> Device Placement :- This determines the strategic position of a device for maintaining security, connectivity & traffic management.

The n/w is divided into 3 separate zones.

LAN (Trusted)		DMZ Boundary Layer		WAN Untrusted
IPs IDs Local Balancer Switches Sensors	Internal Firewall	IPs, IDs, Jump Server Proxy Server Reverse proxy load balancer, sensors	Perimeter Firewall.	Router (ACL)

 Security Zones :- These are the partitions within a network having their own security policies and access control.

The key features of security zones are :-

- i) Segmentation
- ii) Data protection
- iii) Access Control
- iv) Monitoring & logging
- v) Operational efficiency
- vi) Isolation

from cyber attacks (vii) Defence in depth policy.

<2> Attack surfaces: The network attack surface includes all points through which an attacker can potentially gain unauthorized access, manipulate data or disrupt N/w operation.

The key aspects of a n/w surface include the device end point which are vulnerable points for attacker, N/w services like email server & VPN gateways which expose themselves to internet causing cyber attacks, open ports and protocols on network devices generating weak points, phishing operations perform to extract user account and credentials with 3rd party integrations, usage of cloud services all are responsible for exposing the network to cyber attack.

<3> The attack surfaces can be minimized by using methods like vulnerability Assessment, Network segmentation, Single Access point, Access Control with single point of Failure, Security updates & awareness & regular Auditing with MFA (Multi Factor Authentication).

<3> Connectivity:-

<1> It is more than just wires, cables & datapacket and it defines the interaction between devices, systems within a network. It contains wired & wireless connection, cloud integration, remote access & pathways through which information translation happens.

(ii) Connectivity is responsible with the lot of factors like scalability, security, redundancy, complexity & remote work.

(4) Failure Modes:-

This defines how a device/system behaves when it faces a failure or malfunction.

There are 2 modes of operation i.e.

a) Fail-closed mode where the security system defaults to a block state whenever malfunction happens.

b) Fail-open mode where the security system defaults to a completely open state in the presence of failure.

(5) Device Attributes:-

When a network device is responsible for monitoring the data flow & depending upon the functionality & application, devices are categorized as:-

a) Active Device:- which have a pro active force responsible for maintaining security inside a n/w in the presence of potential threat. Ex:- Router

b) Passive Device:- These are observers and they can monitor n/w traffic analyze patterns which will be helpful in preventing potential threat.

c) Inline:- This devices are placed directly in the pathway of n/w traffic helping in processing the data & making real-time decision.

2d) Tap / Monitor:- This devices do not interfere with the flow of n/w traffic, instead they tap into the traffic & duplicate it for analysis and monitoring purpose.

1) Data carries valuable information, which needs to be protected for maintenance of security.

There exist different types of data & similarly data protection formats are also different.

The different types of data are as follows:

a) Regulated Data :-

These data are governed by specific laws & regulation.

It's categories are :-

i) Personally Identifiable Information (PII),

ii) Protected Health Information (PHI),

iii) Financial Data,

iv) Legal data

Corresponding to transactional details & legal processing & consumer data.

b) Intellectual Property :-

These are creative innovations like copyrights, patents & trademarks that has to be highly protected by regulations.

c) Trade Secrets :-

These include confidential business information that provide a competitive advantage & requires technical measures for its protection.

Depending upon readability of data, it can be classified into human readable format & non human readable format.

→ In non-human readable data, it contains binary code, machine language & encrypted data for protecting sensitive information.

Data Classification :-

These serve as the foundations for data protection strategies. Accordingly the classifications are :-

→ Sensitive data - These often refer to privilege information specific to individual.

→ Confidential data - These data are usually related to govt. & research development.

→ Public data - That refers to news & leaflets.

→ Private data - Specific to individual, but not sensitive.

→ Critical & restricted data - It belonging to defence sector.

Data States :-

These defines the condition or the data in which it is residing, & how to utilize it. The categories are :-

(a) data at rest kept inside storage devices,

(b) data in transit that means data travelling across n/w & communication channel.

(c) data in use that is present in random access memory (RAM).

Methods to Secure data:-

The categories are:-

(a) Geographic Restriction:-

These limit data access to user & devices in a specific region only.

(b) Encryption:-

That transfers readable data to unreadable format.

(c) Hashing:-

It converts data into a fixed length string of characters. It is a one way function that can not be reverse engineer.

→ It helps in securely storing passwords in databases to ensure data integrity.

(d) Masking:-

It is a technique to protect sensitive & confidential information by replacing, hiding or scrambling it with fake or anonymize data while protecting the original format or structure.

(e) Tokenization:-

That is a security method where sensitive information is substituted with the random token generated by a tokenization algorithm.

(f) Obfuscation:-

It is a technique used to make the source code or data deliberately complex making it difficult to understand. The two common methods are XOR & ROT13.

{ caesar CT but
with key = 13 }

g) Segmentation & Restricted permission:-

These help in subdividing a network or system into isolated segment & monitoring access control for preventing cyber attack.

→ These are critical concepts in n/w architecture to ensure continuous operation & rapid restoration of services during cyber attacks.

→ Resilience means a capability of withstanding the attack & recovery implies the process of going back to ~~temp~~ normal operation.

→ The objective of network resilience are:-

High Availability (HA),

Fault tolerance,

Redundancy,

Scalability & security

High Availability (HA):-

(1) It is a n/w architecture design approach that ensures system, applications & services remain operational with minimal downtime.

The goal of HA means, continuous access to n/w in the presence of hardware failure, software crashes, maintenance activities and cyber attacks.

(2) HA can be achieved with 2 techniques.

(a) Load Balancer:- It is a device that is used ^{to} evenly distribute incoming n/w traffic across multiple sources & server when there is high volume of traffic incoming.

It uses a virtual IP Address (VIP) to perform the same scheduling algorithms like Least utilized

host, affinity & DNS Round Robin Algorithm.

↓
It helps in optimizing resource ~~optimization~~ utilization.

Affinity:- It is a technique where the load balancer directs request from a client to backend server for optimizing performance.

DNS Round Robin Algorithm:-

The request comes & the load balancer rotates the request across different server evenly.

ii) There are 3 configuration of load balancer.

(a) Active / Active LB:- where every server ^{is} in active mode & takes the load of the n/w.

(b) Active / Passive LB:- where some of the servers are passive without any work & they get converted into active mode whenever the n/w load is high.

(b) Clustering:- It involves grouping multiple servers or nodes together to operate as a single system. It contains an active & passive mode that share a common storage space called quorum disk reinforced by a witness server performing communication like a rhythmic pulse called as heartbeat communication & monitor by a virtual IP.